Rethinking the Illusion of Thinking

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Earlier this year, Apple ignited controversy by publishing "The Illusion of Thinking," prompting heated debate within the AI community. Critics seized upon the findings as conclusive evidence that Large Reasoning Models (LRMs) lack genuine reasoning capabilities, branding them as mere stochastic parrots. Meanwhile, defenders—spearheaded by Lawsen et al. (2025)—fired back, condemning the experimental setup as flawed and the conclusions overstated. We clarify this debate by replicating and refining two of the original study's most contentious benchmarks: Towers of Hanoi and River Crossing. By introducing incremental stepwise prompting and agentic collaborative dialogue, we show that previously reported failures solving the Towers of Hanoi were not purely result of output constraints, but also partly a result of cognition limitations: LRMs still stumble when complexity rises moderately (around 8 disks). Moreover, the River Crossing results initially heralded as catastrophic failures turn out to hinge upon testing unsolvable configurations. Once we limit tests strictly to solvable problems—LRMs effortlessly solve large instances involving over 100 agent pairs. Our findings ultimately defy simplistic narratives: today's LRMs are stochastic, RL-tuned searchers in a discrete state space we barely understand. Real progress in symbolic, long-horizon reasoning demands mapping that terrain through fine-grained ablations like those introduced here.

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I. INTRODUCTION

As Large Reasoning Models (LRMs) advance, reliably assessing their performance has become correspondingly challenging. Benchmark leakage, prompt sensitivity, and high mis-grading rates on complex tasks obscure true capability. Consequently, refining evaluation protocols has become as critical to progress as advances in model architecture [1].

In June 2025, Apple published a highly controversial paper, *The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity* by Shojaee et al. [2]. The authors evaluated the reasoning mechanisms and capabilities of frontier LRMs relying on controllable puzzles rather than standard benchmarks to systematically probe reasoning across varying levels of complexity.

Based on their study, the authors concluded that current LRMs exhibited fundamental limitations in generalizable reasoning. These models consistently failed to perform beyond a certain complexity threshold: they underperformed on high-complexity tasks and tended to overthink in simple ones, where older models—namely Large Language Models (LLMs)—often achieved better results. Their performance also varied significantly across puzzles, successfully executing hundreds of steps in the *Towers of Hanoi* before failing, while rarely completing more than five moves in the *River Crossing* problem.

Another notable finding concerned token usage. LRMs reach peak token consumption when the task is complex but still solvable. Conversely, when the model implicitly detects that the task lies beyond its capabilities, it drastically reduces its output length—suggesting as potential form of internal uncertainty estimation or early abandonment strategy.

The study rapidly attracts attention beyond technical circles and sparks media coverage in outlets such as *The Guardian* and *The Wall Street Journal*, which highlight its warnings about the collapse of LRMs in complex reasoning tasks and raise concerns about the scalability of AI reasoning [3, 4]. In parallel, a number of rapid-response publications and technical commentaries emerge, critically analyzing the study's methodology, results, and conclusions [5, 6].

The study generated a sharp divide within the AI community. On one side, critics interpreted the findings as definitive evidence that LRMs fundamentally lack reasoning capabilities, reducing them to mere stochastic parrots. On the other, defenders argued that the study suffers from flawed benchmarks, questionable experimental design, and overreaching conclusions. The debate quickly transcended academic boundaries. This polarization highlights the urgency of reassessing how we evaluate reasoning in language models, and whether current benchmarks truly reflect cognitive competence or merely expose interface limitations.

Building on this debate, our work emphasizes the influence of task formulation, input validity, and prompt structure on LRM performance. Through our experimental reframing, we observed that structured methodologies—such as stepwise resolution and agentic interaction—can significantly improve results in certain conditions. However, we also identified persistent failure modes that reveal limitations in long-horizon consistency and symbolic generalization. Our analysis suggests that these reasoning breakdowns stem not only from architectural constraints, but also from the inherently stochastic nature of these systems and the optimization methods they rely on. This underscores the need for more nuanced evaluation protocols that go beyond surface-level

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success rates.

It is therefore critical to rigorously characterize these models' limitations, capabilities, and behavioral patterns, and to establish robust evaluation frameworks that can assess their performance in ways aligned with the demands of scientific reasoning [8].

We emphasize that this work does not aim to undermine the contributions of Shojaee et al. [2]. On the contrary, we consider their study both impactful and timely. It opens a valuable discussion on the nature of reasoning in LRMs, and our goal is to complement their findings by introducing alternative perspectives and experimental refinements.

II. FOUNDATIONAL WORK

Aforementioned, Shojaee et al. [2] proposed a set of puzzle-based benchmarks to evaluate the reasoning capabilities of Large Reasoning Models (LRMs). Among these, *Towers of Hanoi* and *River Crossing* (see Fig. 1) emerged as the most discussed and revealing tasks in their study.

Towers of Hanoi is a classic recursive puzzle involving three pegs and a stack of disks of decreasing size placed on one peg in sorted order. The goal is to move the entire stack to a target peg, following two constraints: only one disk may be moved at a time, and no disk may be placed on top of a smaller one. The minimal number of moves required to solve the puzzle grows exponentially with the number of disks, following the expression $2^n - 1$. This makes it a scalable benchmark for evaluating long-horizon planning and symbolic rule-following. Shojaee et al. reported that LRMs handled the initial moves but deteriorated sharply beyond $n \approx 8$: models violated rules, repeated states, or produced invalid sequences.

This trend was interpreted as a symptom of reasoning breakdowns in long-horizon planning. Although the minimal solution path was known and recursive, the models failed to generalize the underlying pattern, especially when prompted to generate full solutions in one pass. Interestingly, when the task exceeded the models' operational limits, the number of tokens used dropped significantly, suggesting that the models implicitly recognized their inability to complete the solution and reduced their effort accordingly. Additionally, the authors found that in simpler configurations, traditional LLMs—without explicit reasoning chains—often outperformed LRMs, as they produced shorter, more direct outputs with fewer computational resources.

River Crossing. In this variant, N agent-actor pairs must cross a river using a boat with limited capacity k, subject to the constraint that no actor may remain with a different agent unless its own agent is present. Difficulty grows with N and depends on k; only instances satisfying $k \geq 4$ or $N \leq 2k - 1$ are solvable [9, 10].

Empirically, in Shojaee et al.'s experiments, LRMs seldom produced more than five valid moves and showed the weakest results across the benchmark. However, these experiments also prompted the model to solve mathematically unsolvable initial configurations, highlighting critical limitations in the evaluation design.

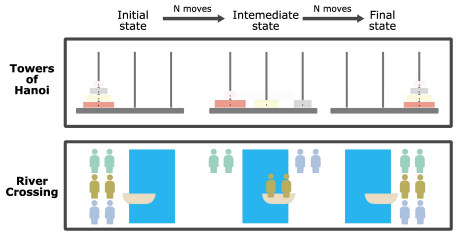

FIG. 1. Illustration of the two benchmark puzzles analyzed in this study: *Towers of Hanoi* (top) and *River Crossing* (bottom). Each panel shows three sequential states: the initial configuration, an intermediate state after a few moves, and the final goal configuration. In Towers of Hanoi, the disks must be moved from the left peg to the right peg following size and move constraints. In River Crossing, each pair of agent and actor (same color) must cross the river while respecting the constraint that no actor can be left with a different agent unless accompanied by their own.

These results were taken as evidence of a deeper limitation: although LRMs displayed emergent reasoning behaviors in some tasks, they lacked the robustness and consistency required to perform reliably across different problem domains.

III. EXPERIMENTAL DESIGN

A. Baseline Design

Our experimental procedures build directly upon the baseline design proposed by Shojaee et al. [2]. To ensure compatibility, we reuse the original prompts (see Code Availability) provided in their study with only minor modifications tailored to our setting.

We employ the Gemini family of models as our LRM, specifically Gemini 2.5 Pro. This model demonstrates state-of-the-art performance in structured reasoning tasks [11–13].

B. Experimental Reframing Strategies

We present alternative methodologies designed to address the limitations observed in the original benchmarks. Each subsection describes a distinct configuration combining task type with specific prompting strategies, aiming to uncover deeper insights into LRM reasoning performance.

1. Stepwise Resolution in Towers of Hanoi:

The original publication shows that as the problem size increases—specifically, as the number of disks in the Towers of Hanoi grows—LRMs become increasingly unable to solve the task. Beyond a certain complexity threshold, models not only fail to generate valid solutions but also tend to produce shorter, incomplete outputs. This reduction in token usage suggests a form of implicit uncertainty management, where the model appears to allocate fewer resources when it "predicts" that the task is likely unsolvable or beyond its capabilities.

One widely held explanation for the LRM's failure in long-horizon tasks like Towers of Hanoi is not a lack of reasoning capacity, but rather a technical limitation. As the number of disks increases, the length of the output required to reach a valid solution grows exponentially, quickly approaching or exceeding the model's maximum output window. Faced with this constraint, the model seems to reduce its effort, generating fewer tokens and often halting early. This behavior suggests that the model anticipates the impracticality of completing the task within the allowed token limit and, rather than wasting resources, implicitly chooses to abandon the attempt. These failures thus reflect constraints on solution length and output budgeting, rather than a fundamental inability to reason [5].

In order to investigate whether token budget limitations are the true cause of systematic failure in large puzzle instances, we adopt a stepwise resolution approach. The overall goal remains the same as in the original experiment, but instead of prompting the LRM to solve the entire problem in a single pass, we divide the task into N subproblems. In each subproblem, the model is asked to generate the next p steps toward the solution, starting from the current configuration. The subsequent prompt then resumes from the final state of the previous iteration. This iterative setup reduces the output burden at each stage, allowing us to test whether performance improves when the model operates under a shorter reasoning horizon.

2. Agentic Dialogue in Towers of Hanoi:

Similar to the previous approach, this procedure also employs stepwise resolution. However, instead of prompting a single LRM with isolated queries to transition between states (separated by p moves), we implement an agentic system composed of two LRMs that collaborate through dialogue. Each agent maintains access to a shared memory and infers the current state based on the latest movements provided by its counterpart. During each turn, the agent receives the last set of moves and is tasked with proposing the next p steps toward the solution.

Recent studies have shown that multi-agent systems offer several advantages over single-model architectures, especially in tasks requiring symbolic reasoning, strategic planning, and coordination. Cooperative agents tend to explore more diverse solution paths, generate higher-quality answers through iterative refinement, and maintain more consistent long-horizon coherence than isolated LRMs [14–18, 21]. These findings support the use of agentic interaction not merely as a heuristic aid, but as a structured enhancement in solving recursively formulated problems like the Towers of Hanoi .

3. River Crossing:

In the original study, this puzzle yields the poorest performance, with LRMs failing to solve any configuration where the number of agent–actor pairs $N > 5$. However, the experimental setup introduces a critical flaw: while increasing N, the boat capacity k remains fixed at 3. Formal solvability criteria for this class of problems establish that a configuration is solvable if and only if it satisfies one of the following conditions [9, 10]:

$$\begin{aligned} &\text{(a)} \quad k \geq 4 \quad \Rightarrow \quad \forall N \geq 1, \text{ the puzzle is solvable,} \\ &\text{(b)} \quad k \leq 3 \quad \Rightarrow \quad N \leq 2k - 1 \quad (\text{i.e., } N \leq 5 \text{ for } k = 3). \end{aligned}$$

By fixing $k = 3$ and increasing N beyond 5, the original experiments include configurations that are unsolvable, invalidating the assessment of model reasoning capabilities in River Crossing tasks.

To ensure a fair evaluation, we replicate the original River Crossing experiments while restricting our tests to configurations that are mathematically solvable under the known constraints. This allows us to isolate reasoning failures from structural impossibility and to more accurately assess the LRM's capacity for symbolic rule-following under valid conditions.

IV. RESULTS

A. Towers of Hanoi

1. Stepwise Resolution:

To evaluate the LRM's ability to solve the Towers of Hanoi puzzle under stepwise prompting, we test configurations of increasing difficulty by varying the number of disks from $N = 3$ to $N = 10$. For each configuration, we run 10 independent trials to obtain robust averages. In this setup, the full puzzle is decomposed into a sequence of substages, with the model instructed to generate a fixed number of p moves toward the solution at each substage.

Fig. 2*a)* illustrates the results. The purple bars indicate the success rate across trials, while the two overlaid curves represent different aspects of token usage. The black curve shows the average total number of tokens